```
    typedef ratio<1, 1000000000000000000> atto;
    typedef ratio<1, 1000000000000000> femto;
    typedef ratio<1, 1000000000000> pico;
    typedef ratio<1, 1000000000> nano;
    typedef ratio<1, 1000000> micro;
    typedef ratio<1, 1000> milli;
    typedef ratio<1, 100> centi;
    typedef ratio<1, 10> deci;
    typedef ratio<10, 1> deca;
    typedef ratio<100, 1> hecto;
    typedef ratio<1000, 1> kilo;
    typedef ratio<1000000, 1> mega;
    typedef ratio<1000000000, 1> giga;
    typedef ratio<1000000000000, 1> tera;
    typedef ratio<1000000000000000, 1> peta;
    typedef ratio<1000000000000000000, 1> exa;
}
```

D.6.1. std::ratio class template

The `std::ratio` class template provides a mechanism for compile-time arithmetic involving rational values such as one half (`std::ratio<1,2>`), two thirds (`std::ratio<2,3>`) or fifteen forty-thirds (`std::ratio<15,43>`). It's used within the C++ Standard Library for specifying the period for instantiating the `std::chrono::duration` class template.

Class definition

```
template <intmax_t N, intmax_t D = 1>
class ratio
{
public:
    typedef ratio<num, den> type;
    static constexpr intmax_t num= see below;
    static constexpr intmax_t den= see below;
};
```

Requirements

`D` may not be zero.